The Impact of Major Injuries on Career Trajectories in Women's Collegiate Lacrosse

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Abstract

This thesis examines the impact of major injuries on the career trajectories of women’s collegiate lacrosse players. Using scraped roster data and injury histories from Atlantic Coast Conference programs, a new dataset was developed to track season-level performance over time. Players were classified into role-based clusters, and an XGBoost model was trained to predict performance metrics season-by-season. Injury simulations compared actual post-injury production to counterfactual projections. Findings reveal that major injuries often lead to substantial declines in performance, though some players exceed expectations. This study highlights the varied consequences of injuries and the challenges of modeling athletic career progression.

Introduction

Injuries have long been a defining factor in the careers of athletes, altering paths, cutting seasons short, and reshaping long-term potential. In collegiate sports, where careers are already short, the effects of major injuries can be particularly profound. Yet for many sports, including women's lacrosse, the full impact of injuries on player development remains underexplored. Limited access to standardized data and inconsistent reporting across programs have left significant gaps in understanding how injuries influence athletic trajectories at the college level.

This thesis aims to address part of that gap by examining how significant injuries affect the careers of women’s collegiate lacrosse players. Through the development and analysis of a new dataset focused on player performance and injury history, this project seeks to better understand the patterns and consequences of injury across different types of players and roles. By combining data collection, player classification, and predictive modeling, the study offers a quantitative lens into the ways injuries shape athletic progression.

Literature Review

Data Scraping

Data scraping is the process of automatically extracting information from websites. It is especially useful for data collection when a centralized or complete data repository does not exist. As Gábor László Hajba[1] explains in *Website Scraping with Python*, data scraping often serves as the foundation for further data analysis projects. Hajba highlights that while scraping is critical, building a robust scraper that can adapt to different website structures and layouts can be challenging.

Manually collecting data from web pages is inefficient and prone to errors, as Ritu Banerjee[2] discusses. Web scraping automates interactions with webpages, offering a faster and more accurate alternative. This automation enables the user to create larger, cleaner datasets for data analysis and machine learning projects.

Clustering

Clustering is the process of grouping data points based on similarity. K-Means is one of the most commonly used clustering algorithms. It partitions data into K groups, where each group is centered around a calculated mean value, known as a centroid. As explained by John Burkardt[3], variance, the squared difference between each data point and the mean, is a key concept in K-Means clustering. When the variance within clusters is low, data points are tightly grouped around their centroids.

Determining the correct number of clusters (K) is crucial for effective clustering. To identify the optimal K, the Elbow Method is often used. As described by Muhammad Ali Syakur et al.[4], the Elbow Method plots the within-cluster sum of squares (WCSS) against different values of K. The "elbow" point, where the rate of decrease sharply slows, suggests the ideal number of clusters to select.

XGBoost

XGBoost (Extreme Gradient Boosting) is a scalable machine learning framework based on gradient boosting, developed by Tianqi Chen and Carlos Guestrin[5] in 2016. Gradient boosting is an ensemble learning technique that builds strong predictive models by combining many weaker learners sequentially.

Lingyu Zhang et al.[6] describe XGBoost as an efficient and scalable implementation of gradient boosting, designed to improve performance by combining weak classifiers in a structured, additive manner.

In a study by Pedro Diniz et al.[7], XGBoost was applied to predict match participation in elite soccer players following Achilles tendon ruptures. Using a classification model, they found that features related to pre-injury match participation were the most important predictors, achieving strong predictive performance.

When applying XGBoost to time series forecasting, special feature engineering is necessary, as Hristos Tyralis et al.[8] explain. Since XGBoost assumes that input features are independent, lagged variables, previous values of the target time series, must be created. These features allow the model to capture the sequential dependencies in time series data, enabling XGBoost to effectively model and forecast temporal patterns.

Data Summary

A significant component of this thesis involved collecting and curating data. Lacrosse analytics can be particularly challenging due to the limited availability of standardized data across programs. To compile the necessary dataset, university webpages were scraped for roster and season-level statistics. Additionally, each player's biography was manually reviewed to identify injury information, specifically noting whether an injury occurred and in which year. Only substantial injuries, defined as those that kept a player out for the majority of a season, were included in this analysis. Unfortunately, most university webpages do not detail the type or severity of injuries, so the analysis focused on a general injury designation.

The dataset was further refined to include only players with at least three seasons of recorded statistics, in order to evaluate career trajectories more comprehensively. Data were gathered from Atlantic Coast Conference (ACC) programs using Sidearm Sports websites, as their consistent formatting made scraping feasible[1]. Schools included in the analysis are Syracuse, Boston College, Louisville, Duke, and the University of North Carolina at Chapel Hill. The dataset spans from the 2015 to the 2024 seasons; 2015 marks one of the earliest years where consistent data are available across programs, with data from 2014 and earlier being less reliable.

At the conclusion of the data collection process, the dataset contained 472 rows of season-level data, with each row representing one season for a player who had at least three seasons recorded. Variables include key performance metrics (e.g., goals, assists, turnovers, games played), physical characteristics (e.g., height), and player roles (e.g., position).

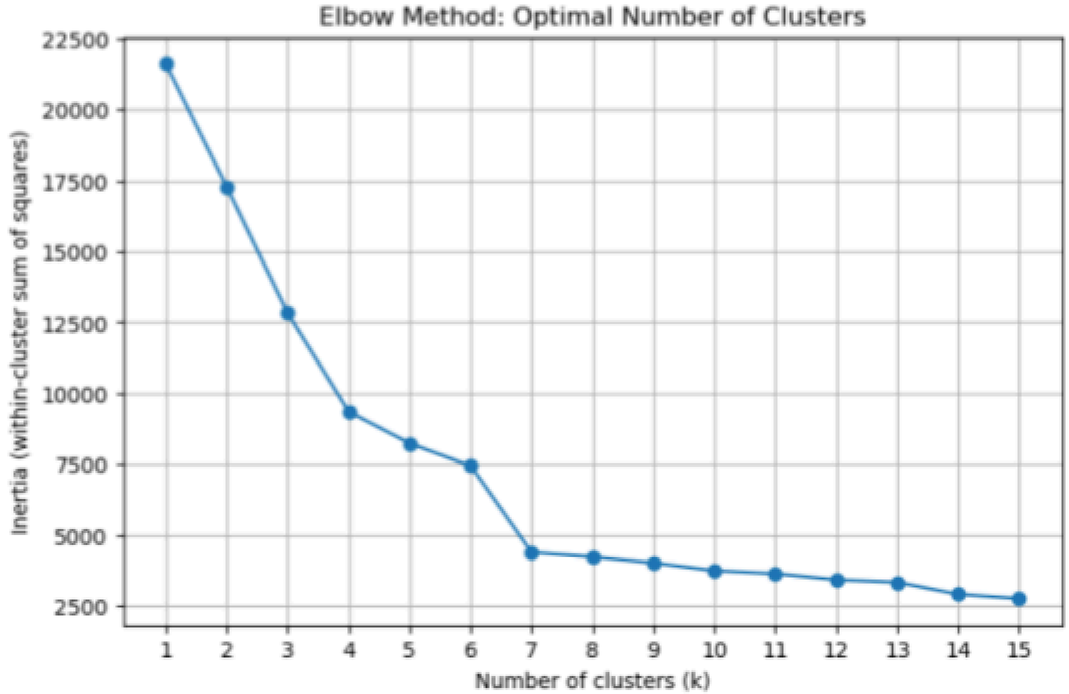
Clustering

To enhance position-based analysis and differentiate between high-impact players and those with limited playing time, a refined player classification system was developed using unsupervised learning. K-Means clustering was applied to group players into distinct roles based on both statistical output and traditional position designations.

K-Means clustering works by partitioning observations into clusters in which each observation belongs to the cluster with the nearest mean (centroid)[3]. Features used for clustering included: position, height, games played percentage (GP%), goals, assists, total shots, shooting percentage, turnovers, ground balls, caused turnovers, free-position shots, and draw controls. To maintain the influence of the traditional position categories, the position variable was given double the weight of other features.

The optimal number of clusters was determined using the elbow method[4], as seen in Figure 1, which involves plotting the within-cluster sum of squares against the number of clusters and identifying the point where the rate of decrease sharply changes. Based on this analysis, seven clusters were selected.

Figure 1: Elbow Method for Optimal Cluster Number

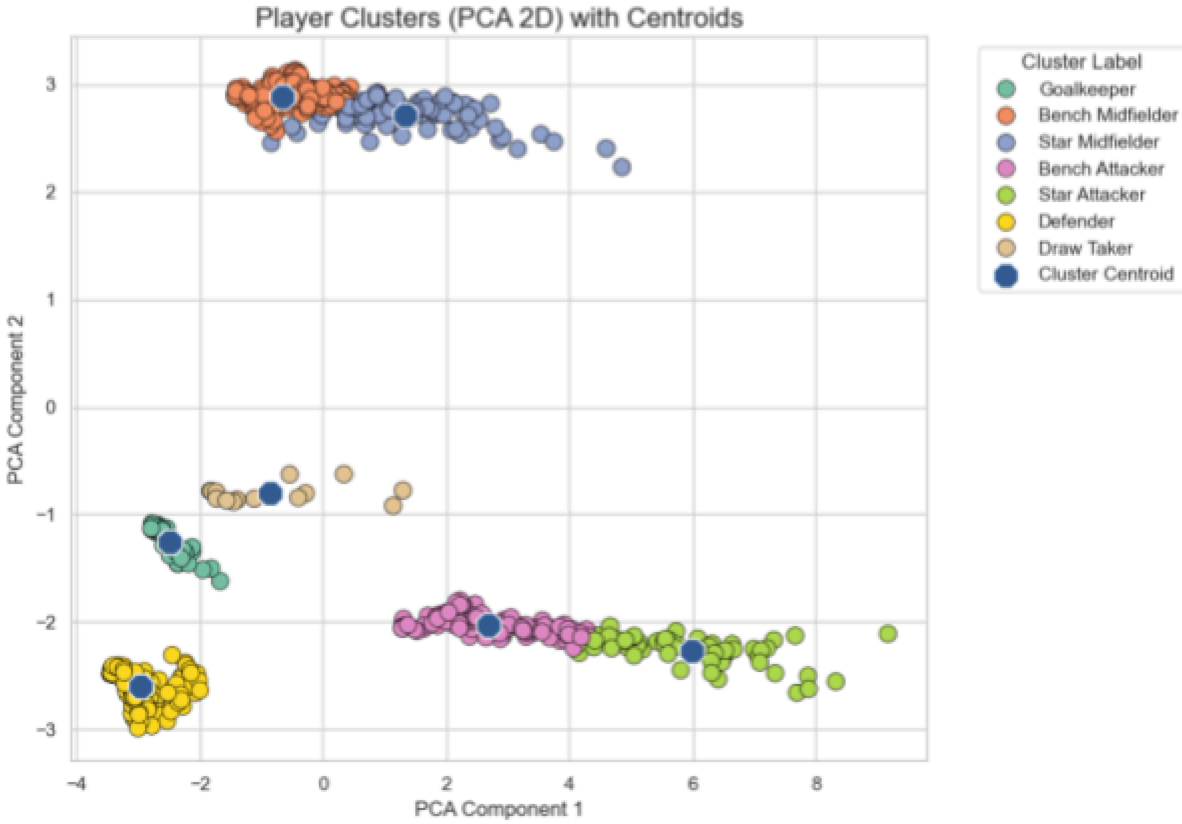


Analysis of the resulting clusters revealed meaningful differentiation as seen in Table 1: goalkeepers, defenders, and draw takers largely remained in distinct clusters, while attackers and midfielders were subdivided into "starters" and "bench" designations. These refined groupings provided a more nuanced classification scheme that better captured role-based differences in career progression. A two-dimensional PCA (Principal Component Analysis) visualization was used to illustrate the cluster separations.

Table 1: Clustering Breakdown

Cluster	Position	Label	Count
0	M	Bench Midfielder	207
1	D	Defender	166
2	GK	Goalkeeper	50
3	A	Bench Attacker	103
4	A	Star Attacker	61
5	M	Star Midfielder	95
6	Draw	Draw Taker	15

Figure 2: Cluster Visualization



The standard position column was subsequently replaced with the clustered position labels in all further analyses, allowing for better examination of how injuries affect players of different on-field roles and impact levels.

Predictive Modeling

To quantify the impact of injuries on player development, a predictive model was constructed using XGBoost, a highly efficient and accurate gradient boosting algorithm. XGBoost builds decision trees sequentially, with each new tree aiming to correct the residual errors of the previous ones, making it well-suited for structured data with complex patterns.[5]

The model aimed to predict season-level performance metrics using per-game statistics, allowing comparisons across teams regardless of schedule length. Additionally, lag features[8] were engineered to capture progression over time by incorporating the previous season's stats for each player (with freshmen assigned zeros for lag values).

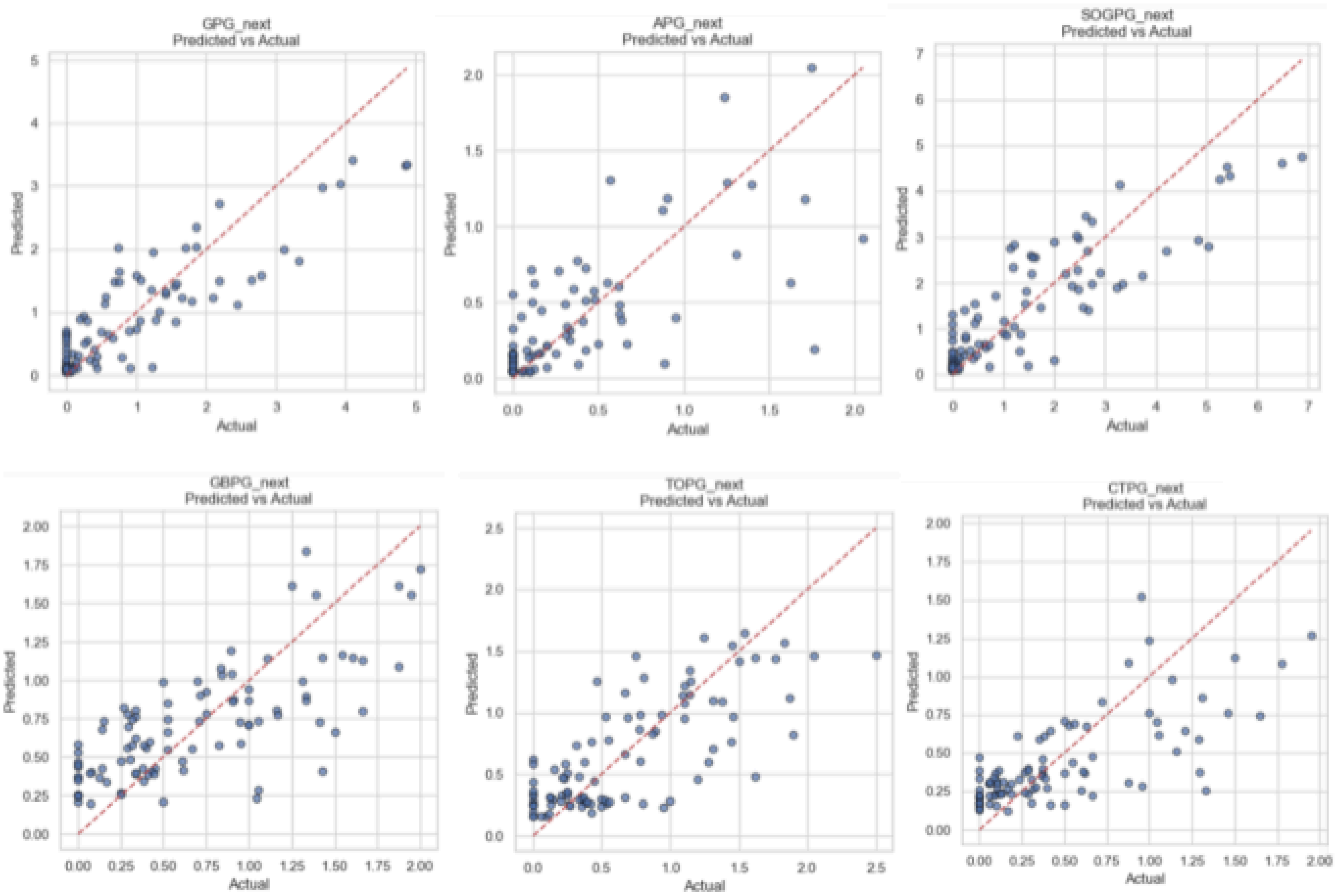
Predictive features included:

- Goals per game (GPG)
- Assists per game (APG)
- Shots on goal per game (SOGPG)
- Ground balls per game (GBPG)
- Turnovers per game (TOPG)
- Caused turnovers per game (CTPG)
- Lagged values of all above statistics
- Games played percentage (GP%)
- Academic year (encoded categorically from freshman through graduate plus redshirts)

GPG and APG features capture basic offensive impact while excluding points per game to reduce multicollinearity. While SOG and GP% represent player usage, CTPG and GBPG express basic defensive impact. Draw controls per game (DCPG) were excluded from the final model because of the lack of predictive power. The lagged version of these features and academic year allows for career trajectory projection.

The model was trained on non-injured player data using an 80/20 train-test split. Since the goal was to predict multiple performance metrics simultaneously, a multi-output regressor was used to wrap XGBoost. This approach trains a separate XGBoost model for each target variable (GPG, APG, SOGPG, GBPG, TOPG, CTPG), enabling comprehensive prediction of future performance.

Figure 3: Predicted Vs. Actual Values by Statistic

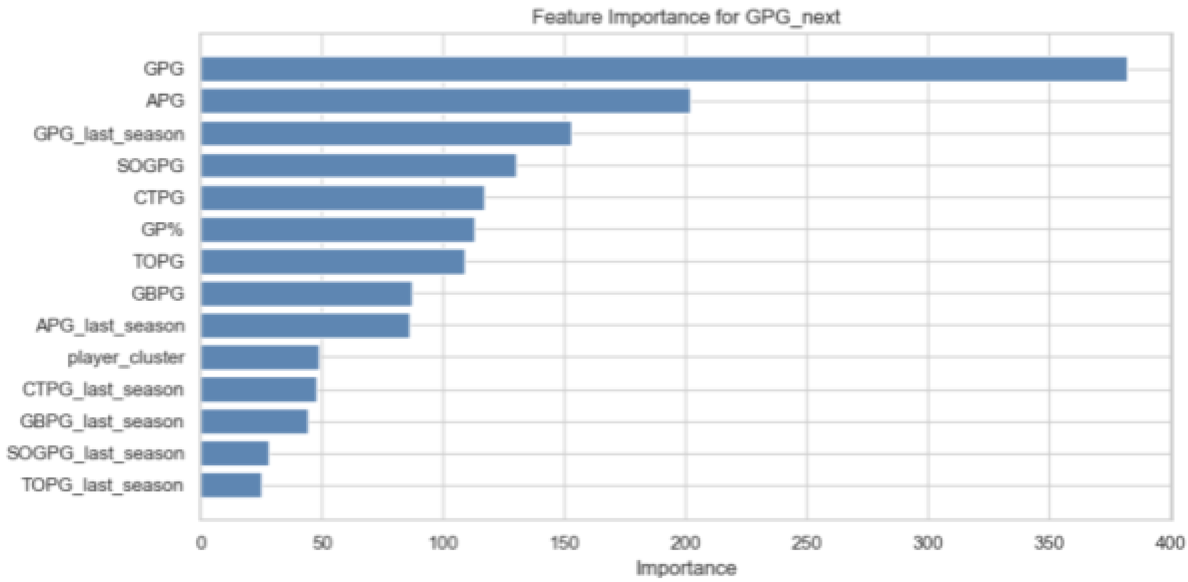


Basic hyperparameter tuning was conducted (including learning rate and number of trees), and the model's accuracy was evaluated as seen in Table 2. Results indicated that GPG was among the most predictable metrics, with its most important predictors being current GPG, APG, and the previous season's GPG. This is seen in the feature importance plot in Figure 4.

Table 2: Model Evaluation

Next Year Stat	MSE	RMSE	R^2
GPG	0.314	0.561	0.761
APG	0.107	0.328	0.524
SOGPG	0.659	0.812	0.751
GBPG	0.138	0.372	0.524
TOPG	0.143	0.379	0.570
CTPG	0.109	0.330	0.514

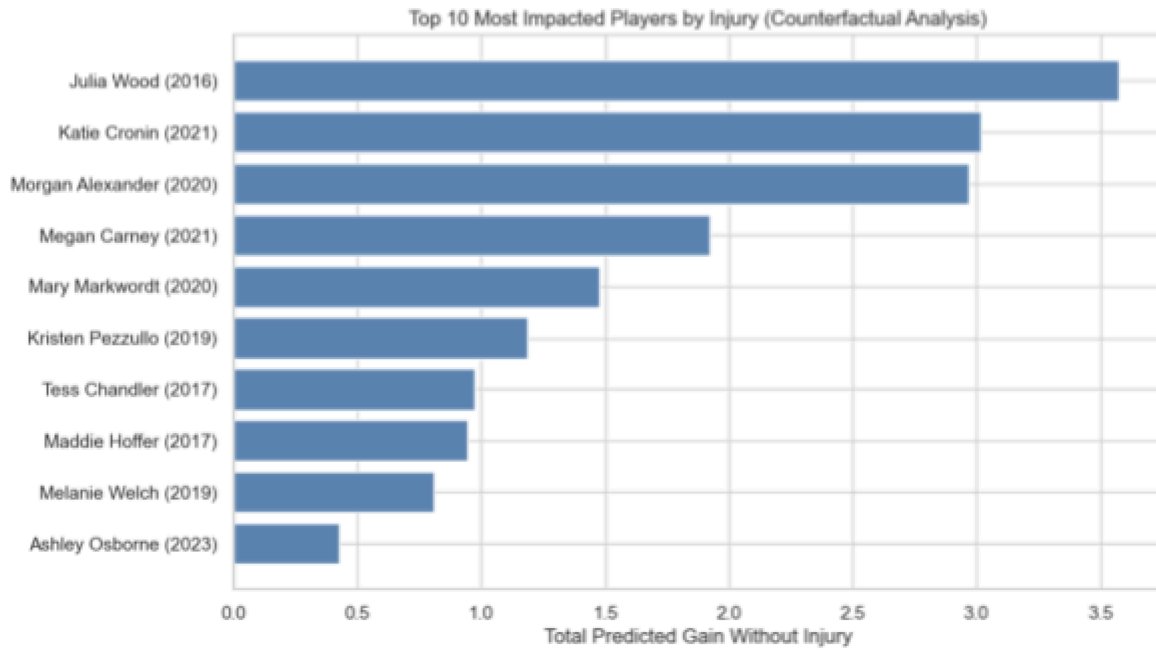
Figure 4: GPG for Following Season Feature Importance



Injury Simulation and Case Studies

Once trained, the model was applied to the injured player subset to simulate hypothetical, injury-free seasons. This involved using the model to generate predicted values for injured seasons, based on lag features from the previous (pre-injury) year.

Figure 5: Single Season Injury Impact Comparing Predicted Season to Actual Following Season



By comparing the predicted and actual total statistics, it was possible to identify which seasons and players were most negatively affected by injuries, as seen in Figure 5. For instance, the 2016 injury sustained by Julia Wood showed a dramatic drop in production as she scored 14 goals in 10 games prior to injury, but only 1 goal in 7 games the following season. This highlights how a major injury can derail a short collegiate career.

Figure 6: Career Differences Across All Stats

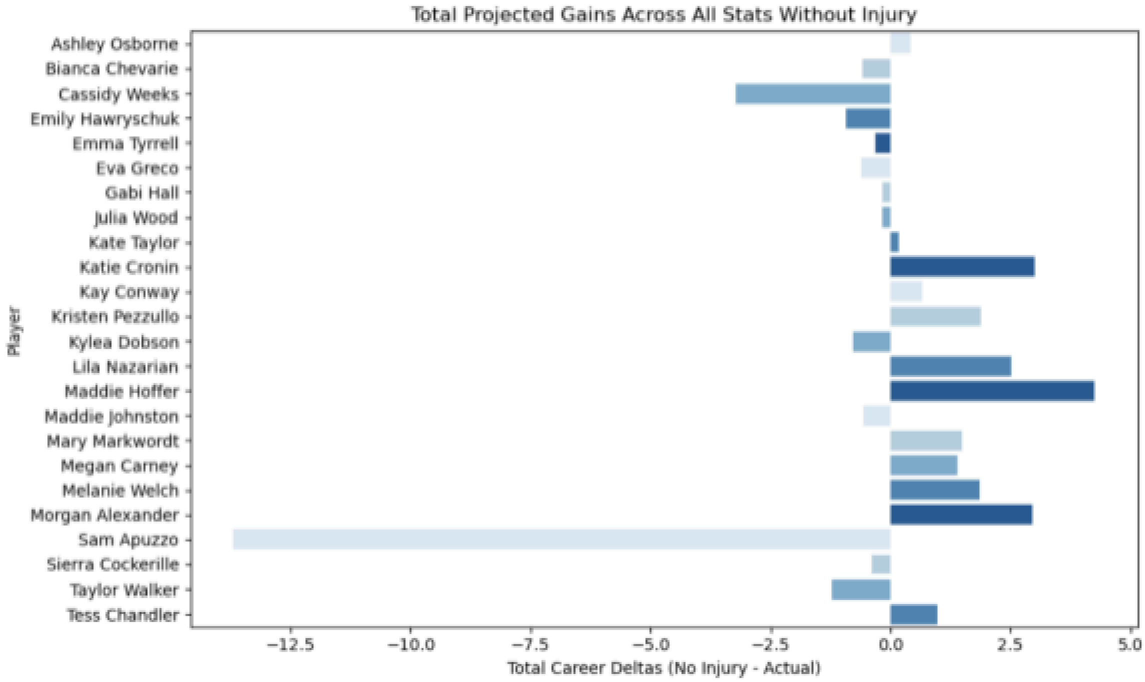
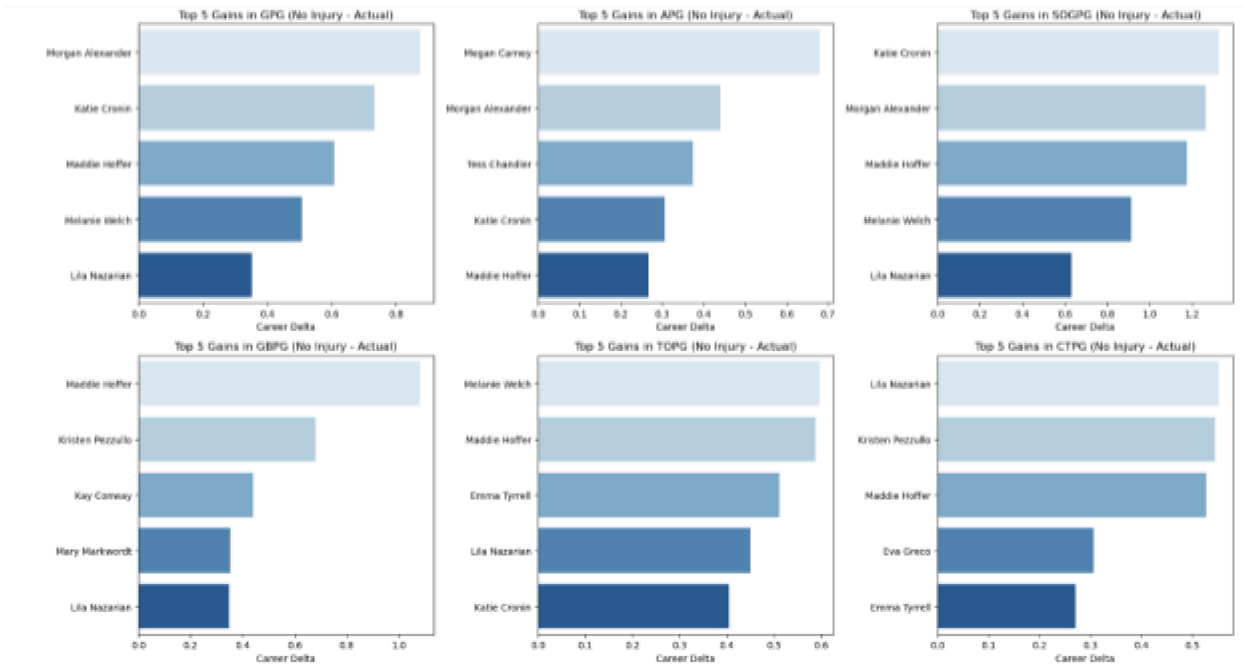


Figure 6 shows the total difference in predicted vs. actual per-game stats across entire careers. While some players, like Boston College's Sam Apuzzo, appeared to outperform model expectations post-injury, such anomalies underscore the limitations of the model. Apuzzo suffered a season-ending injury as a freshman but returned to have one of the most decorated careers in collegiate lacrosse, winning the 2018 Tewaaraton Award and finishing with 278 career goals which puts her in the top 5 all time for this category. These outlier cases are inherently difficult to model due to the rare nature of their post-injury trajectories.

Conversely, players such as Katie Cronin, Maddie Hoffer, and Morgan Alexander exhibited significant declines in production relative to their predicted career trajectories, suggesting injuries may have had lasting negative effects.

Figure 7: Player's Career Differences by Stat



Further breakdowns in Figure 7 show which players were most affected in each individual stat category (e.g., assists, turnovers, ground balls), providing a multidimensional view of injury impact.

Limitations

As mentioned previously, there are some significant model limitations. For one, freshman year injuries result in poor model accuracy, as there is little to no information about the player to make career projections. One way to solve this issue would be to include freshman recruiting rankings in the dataset to account for top players in the class getting injured and having successful careers following. Acquiring these data would be quite difficult, as public women's lacrosse recruiting rankings are not nearly as available or complete as would be necessary. Another limitation is the significance of the injury designation. This analysis focuses on a binary injury variable that does a faulty job of encapsulating different types of injuries and their significance. For example, a player who has a moderate ankle sprain is in the same boat as a player who tears their Achilles. More detailed medical records would be needed to resolve this issue. Additionally, the model is trained on basic stats that do not represent every aspect that goes into a player’s career. There is some omitted variable bias with this model. For example, the model does not know if there is a coaching change, how many minutes a player plays, or how the player treats their recovery. Furthermore, the dataset is fairly limited in size. To get more reliable results, it would be essential to get data from several more programs across both men's and women's lacrosse.

Conclusion

This thesis set out to explore the impact of injuries on women's collegiate lacrosse careers through a combination of data collection, clustering, predictive modeling, and simulation. Despite the inherent challenges in assembling a standardized dataset for lacrosse, the project successfully compiled and analyzed player statistics across five major ACC programs over a ten-year span. By curating a detailed dataset and applying machine learning methods, it became possible to simulate hypothetical, injury-free career trajectories and better understand the extent to which injuries may alter athletic development.

Through the creation of player role clusters and the development of an XGBoost-based predictive model, the analysis demonstrated that injuries often have a measurable negative effect on player outcomes, although the degree of impact varies significantly across individuals. Some players managed to overcome major injuries and achieve elite success, while others saw their career trajectories stall in ways that the model was able to capture and quantify. These findings highlight the unpredictable and individualized nature of injury recovery and athletic performance.

However, this study also brings attention to several important limitations, including the binary treatment of injury severity, limited sample size, and the exclusion of off-field and contextual factors such as coaching changes and rehabilitation quality. Addressing these gaps in future research would allow for a more comprehensive understanding of how injuries shape athletic careers.

Ultimately, this project demonstrates the potential for data-driven approaches to enrich our understanding of athlete development and injury impact in sports like lacrosse, where large-scale analytics are still emerging. Expanding this work to broader datasets and incorporating additional predictive features could provide even deeper insights, ultimately helping programs better support athletes facing career-altering injuries.

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